

What are the top 7 branches of artificial intelligence?

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[Artificial intelligence](#) has made significant strides in a wide variety of industries. Systems that mimic similar behaviors and characteristics found in human intelligence can learn, reason, and understand tasks to take action.

It's important to understand the different concepts in artificial intelligence that help solve real-world problems. This can be done by implementing processes and techniques like [machine learning](#), which is a branch of artificial intelligence.

In this article, we'll go over the main branches of artificial intelligence, such as:

- [Computer vision](#)
- [Fuzzy logic](#)
- [Expert systems](#)
- [Robotics](#)
- [Machine learning](#)
- [Neural networks/deep learning](#)
- [Natural language processing](#)

1. Computer vision

One of the most popular branches of artificial intelligence right now, [computer vision](#), aims to develop techniques that assist computers in seeing and understanding digital images and videos.

Applying machine learning models to images allows computers to identify objects, faces, people, animals, and more. Algorithmic models help computers teach themselves about visual data's contexts, and with enough data fed through a model, computers can teach themselves to distinguish one image from another.

A convolutional neural network works alongside a model to break images down into pixels, giving them tags or labels. The neural network then uses the labels to conduct convolutions, which is a mathematical operation on two functions to produce a third function, and make predictions about what it sees.

Computer vision has applications across industries, such as:

- **Object tracking.** Following or tracking detected objects.
- **Image classification.** An image is classified and accurately predicted to belong to certain classes.
- **Facial recognition.** Face unlock on smartphones unlocks devices by mapping and matching facial features.



2. Fuzzy Logic

Fuzzy logic is a technique that helps to solve issues or statements that can either be true or false. This method copies human decisions by considering

all existing possibilities between digital values of 'yes' and 'no'. Put simply, it measures the degree to which a hypothesis is correct.

You'd use this branch of artificial intelligence to reason about uncertain topics. It's a convenient and flexible way of implementing machine learning techniques and copying human thought logically.

Fuzzy logic's architecture is composed of four parts:

- **Rule base.** Has all the rules and if-then conditions.
- **Fuzzification.** Helps to convert inputs.
- **Inference engine.** Determines the degree of match between rules and fuzzy inputs.
- **Defuzzification.** Converts fuzzy sets into crisp values.

Companies like Nissan use fuzzy logic to control breaks in dangerous situations, which depend on individual car acceleration, speed, and wheel speed.

3. Expert systems

An expert system is a program specializing in a singular task, just like a human expert. These systems are mainly designed to solve intricate problems with human-like decision-making capabilities.

They use a set of rules, called inference rules, that a knowledge base fed by data defines for them. By using if-then logical notions, they can solve complex issues and help in information management, virus detection, loan analysis, and more.

The first expert system was developed in the 1970s, and greatly contributed to the success of artificial intelligence. An example of an expert system is

CaDeT, a diagnostic support system that can help medical professionals by detecting cancer in its early stages.

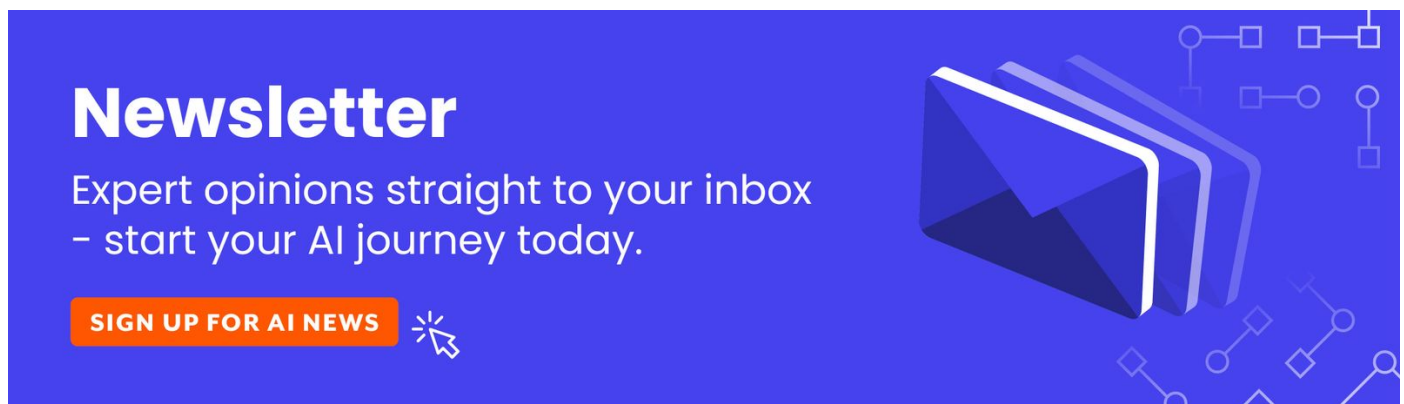
4. Robotics

Robots are programmed machines that can automatically carry out complex series of actions. People control them with external devices, or their control systems can be embedded within themselves.

Robots help humans with tedious and repetitive tasks. AI-powered robots, in particular, can help companies like NASA in [space exploration](#). Humanoid robots are the latest developments and better-known examples of robotic evolution.

Sophia, a robot developed by Hanson Robotics, works through a combination of artificial intelligence and neural networks. She recognizes human faces and understands emotions and gestures – and can even interact with people.

Common examples of robotics in everyday life applications include industries like [manufacturing](#), healthcare, retail, and more.



5. Machine learning

Machine learning is the ability of machines to automatically learn from data and algorithms, and is one of the more demanding branches of artificial intelligence. Machine learning improves performance using past experiences and can make decisions without being specifically programmed to do so.

The process starts with historical data collection, like instructions and direct experience, so that logical models can be built for future inference. Output accuracy depends on data size – a larger amount of data will build a better model, which in turn increases its accuracy.

Machine learning algorithms are classified into three types:

- **Supervised learning.** Machines are trained with labeled data to predict the outcome.
- **Unsupervised learning.** Machines are trained with unlabeled data, with the model extracting information from the input to identify features and patterns, so it can generate an outcome.
- **Reinforcement learning.** Machines learn through trial and error, using feedback to form actions.

6. Neural networks/deep learning

Neural networks are also known as artificial neural networks (ANNs) or simulated neural networks (SNNs). At the heart of [deep learning](#) algorithms, neural networks are inspired by the human brain, and they copy how biological neurons signal to each other.

ANNs have node layers – consisting of an input layer, one or more hidden layers, and an output layer. Each node, also called an artificial neuron, connects to other neurons and has an associated threshold and weight.

When an individual node's output is over a specified threshold value, the

node is activated to send data to the next network layer. Neural networks need training data to both learn and improve accuracy.

7. Natural language processing

Natural language processing allows computers to understand both text and spoken words like humans can. Combining machine learning, linguistics, and deep learning models, computers can process human language in voice or text data to understand the full meaning, intent, and sentiment.

In speech recognition or speech-to-text, for example, voice data is reliably converted to text data. This can be challenging as people speak with varied intonations, emphasis, and accents. Programmers have to teach computers natural language-driven applications so they can understand and recognize data from the beginning.

Some natural language processing use cases are:

- **Virtual chatbots.** They can recognize contextual information to offer customers better responses over time.
- **Spam detection.** Natural language processing text classification can scan language in emails to detect phishing or spam.
- **Sentiment analysis.** Analyzing language used in social media platforms helps to extract emotions and attitudes about products.